

PETRO-QUANT 4

- User Manual

PETRO-QUANT 4 for S8 TIGER 2nd Generation

Original Instructions

The reproduction, transmission or use of this document or its contents is not permitted without express written authority. Offenders will be liable for damages. All rights reserved.

We have checked the contents of this manual for agreement with the hardware and software described. Since deviations cannot be precluded entirely, we cannot guarantee full agreement. However, the data in this manual are reviewed regularly and any necessary corrections are included in subsequent editions. Suggestions for improvement are welcome.

All configurations and specifications are subject to change without notice.

Order no. DOC-M80-EXX163 V2. Updated October 13, 2020.

© 2020 Bruker AXS GmbH, Karlsruhe, Germany.

All trademarks and registered trademarks are the sole property of their respective owners.

Printed in the Federal Republic of Germany.

Bruker AXS GmbH
Östliche Rheinbrückenstr. 49
76187 Karlsruhe, Germany
Tel. +49 721 5 09 97-0
Fax +49 721 5 09 97-56 54
info.baxs@bruker.com
www.bruker.com

Table of Contents

1	Introduction.....	5
2	X-ray Fluorescence Analysis of Liquid Samples.....	6
2.1	Measurements in He Atmosphere	6
2.2	Sample Preparation in Liquid Cups	6
2.3	Compensation of Film Materials	7
3	What will PETRO-QUANT 4 do for you?	8
4	What PETRO-QUANT 4 will not do?	12
5	PETRO-QUANT 4 Sample Preparation	13
5.1	Norms	13
5.2	PETRO-QUANT 4<Be> BASIC	13
6	PETRO-QUANT 4 Validation and Quality Check Procedure.....	15
6.1	PETRO-QUANT 4 Norm Compliant Methods.....	16
6.2	PETRO-QUANT 4<Be> BASIC	18
7	PETRO-QUANT 4 Routine Analysis	19
7.1	PETRO-QUANT 4<Be> BASIC Analysis of Samples with Low Volatility.....	20
7.2	PETRO-QUANT 4<Be> BASIC Analysis of Samples with High Volatility.....	20
7.3	PETRO-QUANT 4<Be> BASIC Analysis of Samples with Oxygen	21
8	PETRO-QUANT 4 Drift Correction.....	22
8.1	Drift Correction (Sensitivity)	23
8.2	Drift Correction (Blank Samples)	24
9	Customizing PETRO-QUANT 4<Be> BASIC.....	25
9.1	Lower Limits of Detection and Counting Times	25
9.2	Customized PETRO-QUANT 4<Be> BASIC Applications	26
9.3	Customized Calibration Methods.....	26
9.4	Additional Elements or Expanded Concentration Ranges.....	27
9.5	Partial Calibrations for Additional Elements.....	28
10	Appendix	29
10.1	Appendix 1: Back-Up and Re-Installation of the PETRO-QUANT 4 Method.....	29
10.2	Appendix 2: Acceptance or Quality Check Troubleshooting.....	29
10.3	Appendix 3: PETRO-QUANT 4<Be> BASIC Reference Samples.....	31
10.4	Appendix 4: Oil Density Determination	34
11	Applications and Drift Correction	35

1 Introduction

PETRO-QUANT is the Bruker AXS solution for the easy analysis of trace elements, wear metals, and additives in all kinds of petrochemical materials. PETRO-QUANT 4 combined with the S8 TIGER II wavelength dispersive XRF spectrometers offers unrivalled accuracy and norm compliant analysis methods. It is a ready-to-analyze package for multi-elemental analysis in various petrochemical or other light matrices right out of the box.

PETRO-QUANT 4 BASIC (successor of OilQuant2 and PETRO-QUANT 2 BASIC) is the Bruker AXS universal calibration for trace analysis in samples of petrochemical matrix (H to F > 95 %). S and Cl can be present in the range from lowest µg/g level up to 5 %. The additive elements Mg, P, Ca, Zn, and Ba are expected to be in a range up to several tenths of a percent. All other 25 elements are in the trace concentration range (less than 1000 µg/g). The calibration and precision parameters are presented in chapter [What will PETRO-QUANT 4 do for you? \[p. 8\]](#).

There are two versions of PETRO-QUANT 4 BASIC:

- The default version which uses primary beam filters only when necessary to improve signal-to-noise ratio or to reduce the measured intensity to within the linear range
- The SampleCare version which uses a 50 µm Be filter for lines which have no filter applied in the default version.

Note



The installation and analysis procedure is the same for both versions. <> are used to indicate where an option is available, e.g. PETRO-QUANT4<Be> BASIC means PETRO-QUANT 4 BASIC or PETRO-QUANT 4Be BASIC.

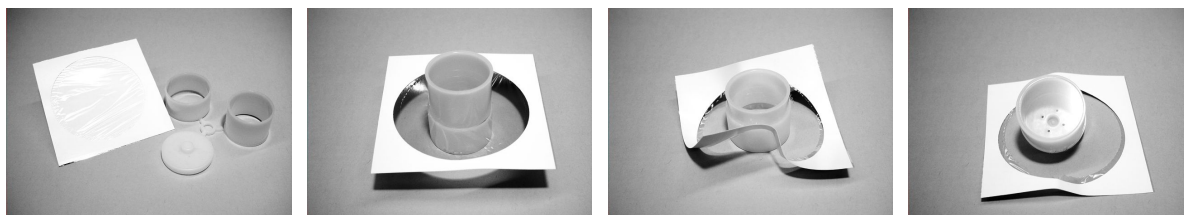
Optionally, PETRO-QUANT 4 offers the following norm compliant turn key solutions:

- ASTM D2622L (S low concentration range; mineral oil matrix)
- ASTM D2622L-D (S low concentration range; Diesel matrix)
- ASTM D2622L-G (S low concentration range; gasoline matrix)
- ASTM D2622M (S mid concentration range; mineral oil matrix)
- ASTM D2622H (S high concentration range; mineral oil matrix)
- ASTM D5059C (Pb in gasoline, low concentrations, part C with internal standard Bi)
- ASTM D6443 (Mg, P, S, Cl, Ca, Cu, Zn low to mid concentration range; lube oil matrix)
- ISO 14596L (S low concentration range; internal standard Zr)
- ISO 14596H (S high concentration range; internal standard Zr)
- ISO 14597 (V low & high concentration range and Ni low concentration range)
- ISO 15597 (Cl and Br low concentration range)
- ISO 20884L (S lowest concentration range)
- ISO 20884H (S low concentration range)
- DIN 13723 (Pb low concentration range; internal standard Bi)
- DIN 51363 (P low to mid concentration range; internal standard Zr)
- DIN 51391Ca (Ca low to mid concentration range; internal standard Sn)
- DIN 51391Zn (Zn low to mid concentration range; internal standard Co)
- DIN 51399 (Additives, wear metals and other contaminants in lubricants)
- DIN 51431 (Mg low to mid concentration range; internal standard Br)
- DIN 51790 (V and Ni low concentration range, S high concentration range; mineral oil matrix)

2 X-ray Fluorescence Analysis of Liquid Samples

When analyzing liquid samples using XRF, there are two important considerations:

- The measurements of liquid samples cannot be performed in vacuum mode but in Helium mode with reduced (samples of low volatility) or atmospheric pressure (samples of high volatility).
- Liquids are analyzed in liquid cups. These are made of two polymer rings and a polymer lid. The bottom of these cups is made of a very thin film.



2.1 Measurements in He Atmosphere

Using a S8 TIGER 2nd Generation spectrometer, two Helium modes are available in SPECTRA^{plus}:

- Helium with reduced pressure of about 150-250 mbar ("He")
- Helium with atmospheric pressure ("HeAtm")

The vapor pressure of the sample is checked and the appropriate mode is defined via the measurement method. The HeAtm mode allows the analysis of highly volatile samples like gasoline. In contrast the reduced He mode consumes less Helium and allows the analysis of low volatile samples like oils.

Note



If liquid samples are analyzed frequently, change the default mode from "Vacuum" to "Helium" or "Atmospheric Helium" in **System Configuration**. The SPECTRA^{plus} software manual will tell you how to do this. This safe mode is selected initially even if the line library is opened directly in **MethodWizard**.

Only opening a measurement method in **MethodWizard** forces the defined mode to be used.

2.2 Sample Preparation in Liquid Cups

The choice of the film material for the liquid cup bottom is essential for the analysis of liquids (and loose powders).

In general, all elements from Na to U can be analyzed in liquids independent from the film material and consequently, the absorption of the fluorescence lines. Unfortunately, the fluorescence lines of lighter elements do not penetrate the film. In these instances, the choice of the film material is always a compromise between low absorption and on the other hand mechanical and chemical stability.

Proven film materials for XRF liquid analysis are:

- 4 µm Prolene® (Prolene is a registered trademark of Chemplex Industries Inc.) film:
 - most often used transparent film and free of contaminations; material of choice if not chemically attacked (e.g. by gasoline samples).
- 3.6 µm Mylar® (Mylar is a registered trademark of DuPont de Nemours Inc.) film:
 - mechanically more stable than Prolene® film, being less transparent and contaminated by P, Ca and Si (see chapter [Compensation of Film Materials](#) [7]); material of choice for gasoline samples.

Note



The appropriate film for unknown liquids can be tested easily. Each sample is poured into two cups being prepared with the different film materials and placed on a paper tissue as seepage warning device. If the Prolene® film is immediately attacked and crinkles (e.g. by gasoline) you can change to Mylar® film. If after hours the liquid cup is lifted and neither the tested film sags nor the paper tissue indicates any stains, experience shows the film material withstands XRF measurements of up to at least 30 min.

2.3 Compensation of Film Materials

Using one universal calibration like PETRO-QUANT 4<Be> BASIC for a big variety of samples requires the application of more than one preparation procedure. To optimize the analysis of traces, it is advantageous to use a blank sample to adjust the calibration offset and to compensate film contaminations (see also chapter [Drift Correction \(Blank Samples\)](#) [▶ 24]).

Please note, using contaminated film materials results in incorrect concentrations of the respective elements. Unfortunately, this kind of contamination cannot be treated like "normal" concentration contaminations since film contaminations are intensity contaminations. Depending on the composition of the sample, the matrix corrected concentration based on the same (film) intensity can be different. The 3.6 µm Mylar® film is contaminated by traces of P, Ca and Si. The corresponding preparation is "PETRO-QUANT4<Be>-Fuel". The compensation of the film contamination must always be related to a single preparation.

The first step is to define a blank reference sample for each respective preparation in the calibration program **ApplicationWizard**. If you need help to do this, please refer to the instructions given in the SPECTRA^{plus} user manual. In Preparation|Contamination, the option "Blank sample" is activated. A new reference sample "BLANK@Preparation name" (in PETRO-QUANT 4<Be> BASIC „BLANK@PETRO-QUANT4<Be>-Oil" is created as well as „BLANK@PETRO-QUANT4<Be>-Fuel"). These additional reference samples have to be defined for the respective lines (in PETRO-QUANT 4<Be> BASIC set for all lines but the Compton line).

A white oil sample is prepared according to the respective preparation (7.0 ± 0.1 g, liquid cup of inner diameter 3.5 cm covered with a 4 µm Prolene® film for PETRO-QUANT4<Be>-Oil or with a 3.6 µm Mylar® film for PETRO-QUANT4<Be>-Fuel). The measurement can easily be defined via the drift correction option of **Loader** (for details see chapter [PETRO-QUANT 4 Drift Correction](#) [▶ 22]). The intensities of the respective elements are transferred to the line library. From now on this blank intensity is automatically subtracted any time the measurement method - preparation combination is applied.

This BLANK measurement should be done anew with any new lot (roll or box) of film material.

Note



As the compensation of the film contamination signal already considers an intensity offset, the calculation of an offset must be disabled when setting up respective calibrations. Although the PETRO-QUANT 4<Be> BASIC is based on standard measurements with a non contaminated film material, the master calibration itself uses a blank correction so that both preparation procedures are handled.

This procedure must be followed for each combination of application and the preparation's respective film material.

3 What will PETRO-QUANT 4 do for you?

PETRO-QUANT 4 is pre-configured to allow “PETRO-analysts” to begin sample measurements quickly and easily. Furthermore, the application long-term maintenance (quality control measurements, drift correction) is reduced to a minimum workload for a routine lab.

The optional norm compliant PETRO-QUANT 4 methods provide:

- Analytical lines and measurement methods set up according to norm requirements.
- Pre-defined procedures for norm compliant calibrations based on the standard samples included in delivery.
- Drift Correction (DC) procedures based on DC sample(s) included in delivery (if required, i.e. for ASTM D2622 (L, L-D, L-G, M, and H), for ASTM D6443, for ISO 20884 (low and high), for DIN 51399, and for DIN 51790).
- Norm compliant Quality Control (QC) procedures on base of QC samples included in delivery.
- Extra Blank Quality Control and corrective procedures for norms analyzing in the lowest concentration range (ASTM D2622 (L, L-D, L-G), ASTM D5059C, ISO 20884, DIN 13723 and DIN 51399).
- Reference samples (standard samples, blank samples and QC samples) already blended with internal standards, i.e. for ASTM D5059C, for ISO 14596 (low and high), for ISO 14597, for ISO 15597, for DIN 13723, for DIN 51363, for DIN 51391 (Ca and Zn), and for DIN 51431.

The PETRO-QUANT 4<Be> BASIC method provides:

- Analytical lines carefully designed to:
 - Use optimum tube and instrument settings.
 - Collect peak and background intensities with a minimum of line overlaps.
 - Use carefully selected Pulse Height Discriminator windows.
- Counting strategy to yield best accuracy within ~17 minutes (S8 TIGER 2nd Generation 4K or 3K) resp. ~23 minutes (S8 TIGER 2nd Generation 1K) measuring time
- Long term Drift Correction by use of well established oil samples
- Instructions for easy and reliable preparation methods to produce safe liquid samples
- Line overlap, background and matrix corrections are automatically applied
- Flexibility for modification of counting times and the usage of PETRO-QUANT4<Be> lines in user defined calibration methods
- Automatic correction for non-infinitely thick samples (sample height lower than critical depth)
- Automatic selection of best line
- Automatic determination of oxygen contents

The PETRO-QUANT 4 BASIC method covers the following list of analytes and respective concentration ranges:

PETRO-QUANT
For S8 TIGER 2nd Generation
User Manual



	Concentration-Range [mg/kg]	Regression-Deviation [mg/kg]	LoD (3 σ ; 12 s) [mg/kg]	LoD (3 σ ; 100 s) [mg/kg]	Avg.Abs.Dev. Range (~30 mg/kg; n=12) [mg/kg]	Avg.Abs.Dev. Range (~100 mg/kg; n=12) [mg/kg]
Na	LoD – 2000	6.6	2.7 (30s)	1.5	5 15	4 14
Mg	LoD – 4000	2.2	0.8 (30s)	0.4	2 8	2 5
Al	LoD – 1250	2.4	1.7	0.6	1 5	2 7
Si	LoD – 2000	2.3	1.9	0.7	1 3	2 8
P	LoD – 5000	1.7	0.5	0.2	0.4 1.2	0.8 2.4
S	LoD – 1000	6.2	0.6	0.2	0.4 1.4	0.6 2.4
S*	LoD – 2.50 %	0.0031 %	~8	~3		
Cl	LoD – 1000	1.3	1.5	0.5	1.3 4.2	1.1 3.6
Cl*	LoD – 2.00 %	0.0012 %	~9	~3		
K	LoD – 2000	1.5	0.5	0.2	0.3 1.1	0.7 2.5
Ca	LoD – 10000	2.7	0.6	0.2	0.4 1.5	0.7 2.2
Ti	LoD – 600	1.7	0.6	0.2	0.4 1.5	0.5 1.2
V	LoD – 600	6.9	2.0	0.7	0.4 1.6	0.6 1.8
Cr	LoD – 600	2.5	0.4	0.1	0.2 0.7	0.3 1.0
Mn	LoD – 600	2.6	0.4	0.1	0.2 0.6	0.3 1.1
Fe	LoD – 2000	2.3	0.4	0.1	0.3 1.1	0.4 1.2
Co	LoD – 600	0.4	0.3	0.1		0.3 1.0
Ni	LoD – 600	1.8	0.2	<0.1	0.2 0.7	0.3 1.0
Cu	LoD – 600	1.5	0.2	<0.1	0.4 1.2	0.3 0.8
Zn	LoD – 2000	2.0	0.1	<0.1	0.1 0.4	0.3 1.0
As K α As K β	LoD – 600	2.1 2.7	0.1 1.0	<0.1 0.3		0.3 1.0 1.0 2.8
Br	LoD – 600	0.8	0.1	<0.1		0.2 0.5
Zr	LoD – 600	1.2	0.4	0.1		0.5 1.9
Mo	LoD – 2000	2.0	0.5	0.2	0.4 1.0	0.4 1.1
Ag	LoD – 600	2.0	1.1	0.4	2 6	1 3
Cd	LoD – 600	1.6	1.3	0.5	1 5	1 3
In	LoD – 600	1.6	1.6	0.6		1 4
Sn	LoD – 600	1.9	1.6	0.6	1 3	2 5
Sb	LoD – 600	2.0	1.9	0.7	1 4	1 4
Ba	LoD – 8000	6.9	2.0	0.7	1.1 4.2	1.3 4.3
W L α W L β	LoD – 1250	0.5 1.5	0.3 0.5	0.1 0.2		0.5 1.7 0.5 2.0

PETRO-QUANT
For S8 TIGER 2nd Generation
User Manual



	Concentration-Range [mg/kg]	Regression-Deviation [mg/kg]	LoD (3 σ ; 12 s) [mg/kg]	LoD (3 σ ; 100 s) [mg/kg]	Avg.Abs.Dev. Range (~30 mg/kg; n=12) [mg/kg]	Avg.Abs.Dev. Range (~100 mg/kg; n=12) [mg/kg]
Tl L α Tl L β	LoD – 600	0.8 1.6	0.2 0.5	<0.1 0.2		0.5 1.5 0.9 2.7
Pb L α Pb L β	LoD – 1250	4.3 1.7	0.2 0.4	<0.1 0.1	0.6 1.8 0.5 1.6	0.4 1.4 0.8 2.2
Bi L α Bi L β	LoD – 600	0.6 1.4	0.2 0.5	<0.1 0.2		0.4 1.3 0.6 1.7

* S and Cl have one line for trace concentrations and one line for minor concentrations

Table 3.1: PETRO-QUANT 4 BASIC Calibration Parameters S8 TIGER 2nd Generation 4K/3K

	Concentration-Range [mg/kg]	Regression-Deviation [mg/kg]	LoD (3 σ ; 12s) [mg/kg]	LoD (3 σ ; 100 s) [mg/kg]	Avg.Abs.Dev. Range (~30 mg/kg; n=12) [mg/kg]	Avg.Abs.Dev. Range (~100 mg/kg; n=12) [mg/kg]
Na	LoD – 2000	9.3	5.0	2.5	4.9 $\pm$ 6.6	6.0 $\pm$ 12.4
Mg	LoD – 4000	4.0	1.7	0.8	3.0 $\pm$ 4.8	2.8 $\pm$ 5.0
Al	LoD – 1250	3.8	1.7	0.8	1.5 $\pm$ 3.0	2.4 $\pm$ 4.0
Si	LoD – 2000	2.6	2.1	0.9	2.2 $\pm$ 3.5	3.0 $\pm$ 5.3
P	LoD – 5000	4.3	0.7	0.3	0.4 $\pm$ 0.9	0.8 $\pm$ 1.6
S	LoD – 1000	1.0	0.9	0.4	0.7 $\pm$ 1.3	1.6 $\pm$ 2.4
S*	LoD – 2.50 %	0.0009 %				
Cl	LoD – 1000	2.0	2.1	0.9	1.0 $\pm$ 1.6	1.0 $\pm$ 1.3
Cl*	LoD – 2.00 %	0.0013 %				
K	LoD – 2000	2.2	0.6	0.3	0.6 $\pm$ 1.0	1.0 $\pm$ 1.5
Ca	LoD – 10000	3.8	0.8	0.4	0.5 $\pm$ 1.0	0.8 $\pm$ 1.7
Ti	LoD – 600	2.2	0.8	0.4	0.3 $\pm$ 0.5	0.6 $\pm$ 1.0
V	LoD – 600	1.7	0.6	0.3	0.6 $\pm$ 0.9	1.0 $\pm$ 1.7
Cr	LoD – 600	2.8	0.5	0.2	0.4 $\pm$ 0.7	0.7 $\pm$ 1.1
Mn	LoD – 600	3.5	0.5	0.2	0.3 $\pm$ 0.5	0.6 $\pm$ 1.2
Fe	LoD – 2000	1.9	0.4	0.2	0.4 $\pm$ 0.6	0.4 $\pm$ 0.7
Co	LoD – 600	0.2	0.3	0.1		0.5 $\pm$ 0.8
Ni	LoD – 600	0.8	0.3	0.1	0.3 $\pm$ 0.7	0.5 $\pm$ 0.9
Cu	LoD – 600	1.6	0.2	0.1	0.3 $\pm$ 0.6	0.5 $\pm$ 0.8
Zn	LoD – 2000	2.7	0.2	0.1	0.3 $\pm$ 0.6	0.5 $\pm$ 1.0

PETRO-QUANT
For S8 TIGER 2nd Generation
User Manual



	Concentration-Range [mg/kg]	Regression-Deviation [mg/kg]	LoD (3 σ ; 12s) [mg/kg]	LoD (3 σ ; 100 s) [mg/kg]	Avg.Abs.Dev. Range (~30 mg/kg; n=12) [mg/kg]	Avg.Abs.Dev. Range (~100 mg/kg; n=12) [mg/kg]
As K α As K β	LoD – 600	0.8 2.5	0.2 1.7	0.1 0.9		0.8 $\pm$ 1.3 1.1 $\pm$ 1.7
Br	LoD – 600	2.3	0.2	0.1		0.5 $\pm$ 0.7
Zr	LoD – 600	0.6	0.6	0.3		0.8 $\pm$ 1.2
Mo	LoD – 2000	1.6	0.8	0.4	0.4 $\pm$ 0.7	0.5 $\pm$ 1.0
Ag	LoD – 600	3.0	2.6	1.2	2.8 $\pm$ 5.4	2.6 $\pm$ 3.9
Cd	LoD – 600	3.3	3.3	1.5	3.2 $\pm$ 5.6	3.7 $\pm$ 6.0
In	LoD – 600	2.1	1.9	0.8		1.5 $\pm$ 2.4
Sn	LoD – 600	3.8	2.0	0.9	0.8 $\pm$ 1.2	1.7 $\pm$ 2.6
Sb	LoD – 600	2.0	2.4	1.1	1.4 $\pm$ 2.3	1.8 $\pm$ 3.2
Ba	LoD – 8000	4.2	2.4	1.1	1.2 $\pm$ 2.1	1.0 $\pm$ 1.9
W L α W L β	LoD – 1250	0.7 1.0	0.5 0.7	0.2 0.3		1.2 $\pm$ 2.0 0.6 $\pm$ 1.0
Ti L α Ti L β	LoD – 600	0.6 1.8	0.3 0.8	0.1 0.4		0.7 $\pm$ 1.0 1.6 $\pm$ 2.9
Pb L α Pb L β	LoD – 1250	1.4 1.4	0.3 0.8	0.1 0.4	1.3 $\pm$ 2.0 1.0 $\pm$ 1.7	0.7 $\pm$ 1.2 1.7 $\pm$ 2.9
Bi L α Bi L β	LoD – 600	0.4 1.4	0.4 0.9	0.2 0.4		0.7 $\pm$ 1.1 1.2 $\pm$ 2.5
* S and Cl have one line for trace concentrations and one line for minor concentrations.						

Table 3.2: PETRO-QUANT 4 BASIC Calibration Parameters S8 TIGER 2nd Generation 1K

Additional measurement time is spent on several background positions to determine accurate net intensities for each element. Thus, the default measurement method will produce data for all of these analytes within ~17 (4 kW and 3 kW) or ~23 (1 kW) minutes measuring time.

4 What PETRO-QUANT 4 will not do?

PETRO-QUANT 4 is designed to analyze traces, minors and majors in a wide variety of petrochemical matrices. It is not intended to analyze:

- Samples that have significant amounts of elements not listed in the two tables above (see chapter [Additional Elements or Expanded Concentration Ranges \[▶ 27\]](#)).
- Samples that have concentration ranges clearly above those listed in the two tables above (e.g. additive concentrates; but see chapter [Additional Elements or Expanded Concentration Ranges \[▶ 27\]](#)).

X-ray fluorescence is a comparative technique. PETRO-QUANT 4 is based on homogeneous samples with a flat film at the bottom of the liquid cup (no sagging). To aim for best performance you should ensure that your liquid samples fulfill these requirements. Three procedures (outlined in chapter [PETRO-QUANT 4 Sample Preparation \[▶ 13\]](#)) and their preparation materials are provided to help you creating good and safe liquid samples.

When analyzing samples by PETRO-QUANT 4, it is important to note that the software, method and spectrometer are not able to correct for the following sample effects:

- Non-homogeneous samples
- Segregating wear metals
- Sagging sample cups films

The PETRO-QUANT 4<Be> BASIC measurement parameters are not designed for the lowest Limit of Detection (LoD) and minimum measurement times for the analysis of one specific sample type. Measurement methods and parameters are optimized for a universal multi-elemental analysis system for all kinds of petrochemical samples. Therefore, the measurement time per element as well as number and position of background positions can be optimized for specific sample types (see chapter [Customizing PETRO-QUANT 4<Be> BASIC \[▶ 25\]](#)).

5 PETRO-QUANT 4 Sample Preparation

All PETRO-QUANT 4 methods, i.e. all installed norm compliant methods and PETRO-QUANT 4<Be> BASIC use just three different preparation procedures, one for oils (norms and PETRO-QUANT4<Be> Basic) and two for fuels (one for norms and one for PETRO-QUANT4<Be> Basic).

5.1 Norms

According to chapter [Sample Preparation in Liquid Cups](#) [▶ 6] there are two kinds of film material to prepare the liquid cups. Furthermore, most of the samples requiring the Mylar® film preparation are of high volatility. Therefore, the sample mass used for this preparation technique is higher than for the oil samples' preparation procedure.

Oil Samples

The liquid cups are prepared using Prolene® film of 4 µm thickness. At any one time 7.0 ± 0.1 g are weighed before each single measurement. The respective cup is placed 30 s on a non-contaminating underlay for leakage testing.

The respective preparations are:

ASTM-D2622L | ASTM-D2622M | ASTM-D2622H | ASTM-D6443 | DIN51363 | DIN51391Ca | DIN51391Zn | DIN51399 | DIN51431 | ISO14596L | ISO14596H | ISO14597 | ISO15597

As blank sample compensation is related to a specific combination of preparation and application, there are two preparations that allow an offset compensation by a blank sample: ASTM-D2622L, and DIN51399 (for details see chapter [Drift Correction \(Blank Samples\)](#) [▶ 24]).

Fuel Samples

The liquid cups are prepared using Mylar® film of 3.6 µm thickness. At any one time 10.0 ± 0.1 g are weighed before each single measurement. The respective cup is placed 30 s on a non-contaminating underlay for leakage testing.

The respective preparations are:

ASTM-D2622L-D | ASTM-D2622L-G | ASTM-D5059C | DIN13723 | DIN51790 | ISO20884

As blank sample compensation is related to a specific combination of preparation and application, there are five preparations that allow an offset compensation by a blank sample: ASTM-D2622L-D, ASTM-D2622L-G, ASTM-D5059C, DIN13723, and ISO20884 (for details see chapter [Drift Correction \(Blank Samples\)](#) [▶ 24]).

5.2 PETRO-QUANT 4<Be> BASIC

PETRO-QUANT 4<Be> BASIC is the universal calibration for all kinds of oils and fuels. Thus, it requires a drift/mode correction system that allows running highly volatile samples with a calibration that has been set up with a different preparation/mode. The calibration reference is the application of a helium atmosphere of reduced pressure and the preparation "PETRO-QUANT4<Be>-Oil" (see below). SPECTRA^{plus} conveniently allows running samples prepared in different ways and measured in different modes by preparation definitions in the database and a few drift correction measurements. So a second preparation for PETRO-QUANT 4<Be> BASIC analyses has been set up (see below).

PETRO-QUANT4<Be>-Oil

The liquid cups are prepared using Prolene® film of 4 µm thickness. At any one time 7.0 ± 0.1 g are weighed before each single measurement. The respective cup is placed 30 s on a non-contaminating underlay for leakage testing.

PETRO-QUANT4<Be>-Fuel

The liquid cups are prepared using Mylar® film of 3.6 µm thickness. At any one time 7.0 ± 0.1 g are weighed before each single measurement. The respective cup is placed 30 s on a non-contaminating underlay for leakage testing.

Due to the most probably limited amount of sample the mass is reduced to 7.0 g compared to the norm preparation procedure.

Meniscus Correction

Both preparations allow the definition of a density and a so called meniscus to achieve optimal results for a big variety of sample types, different sample amounts, and variable oxygen contents (for details see chapter [PETRO-QUANT 4 Routine Analysis](#) ► 19]).

The meniscus correction considers that the surface of any oil or fuel sample is not flat but lens-like. Especially when preparing small sample amounts, it becomes evident that the sample thickness at the liquid cup walls is bigger than that in the middle of the cup. As simple fundamental parameter software packages always assume samples with flat surfaces, deviations from good results are inevitable. The advanced handling of sample shapes in SPECTRA^{plus} allows compensating for this effect. A value of -1 mm has turned out to be the best universal setting for the analysis of various oils and fuels.

6 PETRO-QUANT 4 Validation and Quality Check Procedure

The PETRO-QUANT 4<Be> BASIC method is installed, tested and validated on your S8 TIGER 2nd Generation before it leaves the factory. All the associated files and measurements are loaded onto your system PC. A drift and blank correction have to be carried out after installation for oils and fuels and PETRO-QUANT 4<Be> BASIC is ready to use (for details see chapter [PETRO-QUANT 4 Drift Correction \[▶ 22\]](#)). The drift correction compensates for any small changes in intensity that may have occurred following transportation and installation of your system.

PETRO-QUANT 4 norm compliant methods, that apply an internal standard, do not need drift correction measurements at all as the internal standard reference compensates intensity losses as well as matrix differences. Thus, the methods ASTM-D5059C, DIN13723, DIN51363, DIN51391Ca/Zn, DIN51431, ISO14596L/H, ISO14597, and ISO15597 must not be drift corrected.

As the lower concentrations of the scope of ASTM-D5059C and DIN13723 are close to the limit of determination, there is a blank correction for the lead line (if you are in doubt about the difference between drift and blank correction, please consult your SPECTRA^{plus} manual and chapter [PETRO-QUANT 4 Drift Correction \[▶ 22\]](#)). Further norm compliant methods that do require a drift and blank correction (due to a scope close to the limit of determination) are ASTM-D2622L, ASTM-D2622L-D, ASTM-D2622L-G, DIN51399, ISO20884L, and ISO20884H.

Norm compliant methods that do require a drift correction only are ASTM-D2622M, ASTM-D2622H, ASTM-D6443, and DIN51790.

Once the Drift Correction samples have been measured, the PETRO-QUANT 4 methods are ready for your petrochemical samples.

However, to ensure good data quality and to validate the method(s), Quality Check (QC) samples are provided. These samples have known values for all PETRO-QUANT 4 analytes and are used as check samples to validate the installation of PETRO-QUANT 4 on your S8 TIGER 2nd Generation.

Note



Please note that the values used in the acceptance procedure are 'consensus' values. They are presented by the issuing body with a range of estimated error and notes of caution regarding accuracy. The analytical tolerances quoted in the acceptance procedure take into account the estimated errors of the standard values, analytical error and sample preparation error.

PETRO-QUANT 4 Validation

The PETRO-QUANT 4 Analytical Acceptance Test documents provide an Acceptance Certificate which includes tables showing consensus values, allowed tolerances, factory derived results and a space to enter your measured values collected from PETRO-QUANT 4 methods. The tolerance values allow for analytical error, preparation error and the reported uncertainty of the consensus values. The respective tables need completing, checking and validating. If the data is within the allowed tolerances then the installation has been successful and the Acceptance Certificates can be signed and stored with the system documentation.

PETRO-QUANT 4 Quality Check

All PETRO-QUANT 4 QC samples for long term checks are included in the specification database of SPECTRA^{plus}. The screen display on the external computer as well as on the touchscreen indicates the specification compliance by color. A green marker indicates that the result is within the tolerances defined in the Acceptance Protocol. Any mismatch is pointed out by red color.

If it is not possible to achieve results within the allowed tolerances please refer to the troubleshooting chapter [Appendix 2: Acceptance or Quality Check Troubleshooting \[▶ 29\]](#) of this manual.

6.1 PETRO-QUANT 4 Norm Compliant Methods

Analyzing PETRO-QUANT 4 QC samples is quick and straightforward.

Method	Preparation	QC samples	Application	Results Format
ASTM-D2622L	7.0 g 4 µm Prolene®	QCBlank_ASTM-D2622L QCL_ASTM-D2622L QCH_ASTM-D2622L	ASTM-D2622L	S low
ASTM-D2622L-D	10.0 g 3.6 µm Mylar®	QCBlank_ASTM-D2622L-D QCL_ASTM-D2622L-D QCH_ASTM-D2622L-D	ASTM-D2622L-D	S low
ASTM-D2622L-G	10.0 g 3.6 µm Mylar®	QCBlank_ASTM-D2622L-G QCL_ASTM-D2622L-G QCH_ASTM-D2622L-G	ASTM-D2622L-G	S low
ASTM-D2622M	7.0 g 4 µm Prolene®	QC_ASTM-D2622M	ASTM-D2622M	S high
ASTM-D2622H	7.0 g 4 µm Prolene®	QC_ASTM-D2622H	ASTM-D2622H	S high
ASTM-D5059C	10.0 g 3.6 µm Mylar®	QCBlank_ASTM-D5059C QCL_ASTM-D5059C QCH_ASTM-D5059C	ASTM-D5059C	ASTM-D5059C
ASTM-D6443	7.0 g 4 µm Prolene®	QC_ASTM-D6443	ASTM-D6443	ASTM-D6443
DIN13723	10.0 g 3.6 µm Mylar®	QCBlank_DIN13723 QCL_DIN13723 QCH_DIN13723	DIN13723	DIN13723
DIN51363	7.0 g 4 µm Prolene®	QCL_DIN51363 QCH_DIN51363	DIN51363	DIN51363
DIN51391Ca	7.0 g 4 µm Prolene®	QC_DIN51391Ca	DIN51391Ca	DIN51391Ca
DIN51391Zn	7.0 g 4 µm Prolene®	QC_DIN51391Zn	DIN51391Zn	DIN51391Zn
DIN513999	7.0 g 4 µm Prolene®	QCBlank_DIN51399 QC_DIN51399	DIN51399	DIN51399
DIN51431	7.0 g 4 µm Prolene®	QC_DIN51431	DIN51431	DIN51431
DIN51790	10.0 g 3.6 µm Mylar®	QC_DIN51790	DIN51790	DIN51790
ISO14596L	7.0 g 4 µm Prolene®	QCL_ISO14596L QCH_ISO14596L	ISO14596high	S high
ISO14596H	7.0 g 4 µm Prolene®	QCL_ISO14596H QCH_ISO14596H	ISO14596low	S high

Method	Preparation	QC samples	Application	Results Format
ISO14597	7.0 g 4 µm Prolene®	QCL_ISO14597L-V QCH_ISO14597L-V QCL_ISO14597H-V QCH_ISO14597H-V QCL_ISO14597-Ni QCH_ISO14597-Ni	ISO14597	ISO14597
ISO15597	7.0 g 4 µm Prolene®	QCL_ISO15597-Br QCH_ISO15597-Br QCL_ISO15597-Cl QCH_ISO15597-Cl	ISO15597	ISO15597
ISO20884L	7.0 g 4 µm Prolene®	QCL_ISO20884L QCH_ISO20884L	ISO20884L	S low
ISO20884H		QCL_ISO20884H QCH_ISO20884H	ISO20884H	S low

Table 6.1: PETRO-QUANT 4 Quality Check Samples' Measurement

To measure the norm check samples (preparation see [Table 6.1 \[16\]](#) above) place them in steel sample holders fitted with 34 mm aperture steel masks. Ensure that your S8 TIGER is fitted with a 34 mm collimator mask.

Place the sample holders in the instrument loader and start PETRO-QUANT 4 reference samples' measurements, option "Control Samples". Please refer to the SPECTRA^{plus} user manual to understand how to do this. When the results have to be displayed on the screen of the external computer, the respective result windows (see [Table 6.1 \[16\]](#) above) can be opened via [RESULTS] of the SPECTRA^{plus} **Launcher**. Once the samples have been measured, the results are available.

Each of the QC-samples has consensus values for PETRO-QUANT 4 analytes. To validate the installation it is necessary to compare the measured values against the consensus values.

6.2 PETRO-QUANT 4<Be> BASIC

Analyzing PETRO-QUANT 4<Be> BASIC QC samples requires a little modification.

Method	Preparation	QC samples / Application	Results Format
PETRO-QUANT 4<Be> BASIC	PETRO-QUANT4<Be>-Oil i.e. sample mass 7.0 ± 0.1 g liquid cup film 4 µm Prolene®	QC<Be>-Paraffin PETRO-QUANT4<Be>	PETRO-QUANT4<Be>
		QC<Be>-CON S21&K&Sb 100 ppm	
		QC<Be>-XRS1 PETRO-QUANT4<Be>	
		QC<Be>-XRS2 PETRO-QUANT4<Be>	

Table 6.2: PETRO-QUANT 4<Be> BASIC Quality Check Samples' Measurement

Note



The CONOSTAN reference sample bottles for DC and QC are labelled as "S-21+K+Sb". The "+" character is not a valid character in SPECTRAplus names. Therefore, wherever this sample is referenced in the software, it is called "S21&K&Sb".

As the QC samples of the universal PETRO-QUANT 4<Be> BASIC all have different densities and oxygen contents, the direct start via the drift correction procedure of SPECTRA^{plus} leads to suboptimal results. In order to define the correct application and to input the correct preparation parameters, the QC samples are started via a sample list called "PETRO-QUANT4<Be> Oil QC.txt" from an input format called "PETRO-QUANT4<Be>.def" (if you are in doubt on how to work with sample lists, the SPECTRA^{plus} software manual will help you). The respective applications and preparation procedure are given in [Table 6.2 \[18\]](#) above.

To measure the oil check samples (preparation [Table 6.2 \[18\]](#) above) place them in steel sample holders fitted with 34 mm aperture steel masks. Ensure that your S8 TIGER is fitted with a 34 mm collimator mask.

Each of the QC-samples has consensus values for PETRO-QUANT 4<Be> BASIC analytes. To validate the installation it is necessary to compare the measured values against the consensus values.

7 PETRO-QUANT 4 Routine Analysis

The norm compliant routine analysis of samples is convenient and straightforward. After opening of the respective result window and activation of the routine def files, in case of external computer usage, the routine analysis via **TouchControl™** or the SPECTRA^{plus} **Loader** program of the external computer just requires the input of the samples' name and the selection of the method. The SPECTRA^{plus} manual or your training documents will remind you how to do this.

PETRO-QUANT 4<Be> BASIC can be used for direct analysis (see chapter [PETRO-QUANT 4<Be> BASIC Analysis of Samples with Low Volatility \[20\]](#) and [PETRO-QUANT 4<Be> BASIC Analysis of Samples with High Volatility \[20\]](#)) as well as for customer tailored methods and calibrations (see chapter [Customizing PETRO-QUANT 4<Be> BASIC \[25\]](#)).

All applications can be started from the sample def file "PETRO-QUANT4<Be>.def" (for details see chapter [PETRO-QUANT 4<Be> BASIC Analysis of Samples with Low Volatility \[20\]](#) and [PETRO-QUANT 4<Be> BASIC Analysis of Samples with High Volatility \[20\]](#)). This file allows compensation of different densities ("Density" column) and sample amounts ("total-g" column).

For a direct measurement, predefined applications for "standardless" analysis can be used. In the external computer's loader program two basic applications ("PETRO-QUANT4<Be>-Oil" and "PETRO-QUANT4<Be>-Fuel" can be started via the sample def file "_Routine-He.def".



Note

The PETRO-QUANT 4<Be> BASIC default preparations use a density of 0.87 g/cm³ in the preparation "PETRO-QUANT4<Be>-Oil" and 0.74 g/cm³ in the preparation "PETRO-QUANT4<Be>-Fuel".

Density and sample mass defined by this def file have the following default entries that can be overwritten:

Density (~Oil)	0.87 g/cm ³
Density (~Fuel)	0.74 g/cm ³
Sample mass	7.0 g

The diameter of the sample cups is fixed by the def file.

If the unknown sample has a considerably different density, PETRO-QUANT 4<Be> BASIC allows a simple measurement and modification of the density. Thus, it includes a balance and disposable cups (volume 3.81 cm³ when filled to a flat surface at the top of the cup, i.e. no meniscus) to determine the respective density according the following formula:

$$\rho \text{ [g/cm}^3\text{]} = m \text{ [g]} / V \text{ [cm}^3\text{]} \text{ respectively } \rho \text{ [g/cm}^3\text{]} = m \text{ [g]} / 3.81 \text{ cm}^3$$

See [Appendix 3: PETRO-QUANT 4<Be> BASIC Reference Samples \[31\]](#) for a table of sample mass values and the respective densities.

PETRO-QUANT 4<Be> BASIC also assists in creating specialist methods or calibrations, offering lines (pre-calibrated sets of measurement parameters) that are optimized for trace analysis in a light matrix and are already drift corrected (see [Customizing PETRO-QUANT 4<Be> BASIC \[25\]](#)).

7.1 PETRO-QUANT 4<Be> BASIC Analysis of Samples with Low Volatility

- Samples having a low volatility are typically analyzed by using the Helium mode of reduced pressure ("He") and Prolene® film. PETRO-QUANT 4<Be> BASIC offers 4 applications for this. "PETRO-QUANT4<Be> Oil" defines CH₂ as matrix compound (for samples with considerable oxygen contents please see chapter [PETRO-QUANT 4<Be> BASIC Analysis of Samples with Oxygen \[21\]](#)). In combination with SPECTRA^{plus} version 2.4.0.4 or higher an automatic line selection allows to run all kinds of samples with one application. There are different elements that have two lines in the default PETRO-QUANT4<Be> BASIC measurement method:
- S and Cl: one line for trace analysis (KA1-Tr/PETRO-QUANT4<Be>) and one line for minor analysis (KA1-Min/PETRO-QUANT4<Be>)
- As: two K lines (KA1/PETRO-QUANT4<Be> and KB1/PETRO-QUANT4<Be>)
- W, Tl, Pb, and Bi: two L lines (LA1/PETRO-QUANT4<Be> and LB1/PETRO-QUANT4<Be>)

In case of lines for different concentration ranges, SPECTRA^{plus} 2.4.0.4 or higher allows to define an intensity threshold to select the best line. The trace lines are set as default in the respective evaluation model. When the trace region's intensity limit is exceeded, SPECTRA^{plus} automatically activates the alternate –Min (minor) line for evaluation.

In case of different lines (KA1 and KB1 or respectively LA1 and LB1) set up to avoid severe line overlaps, SPECTRA^{plus} allows to define a maximum overlapped intensity rate.

One of the two lines is set as default in the respective evaluation model (e.g. As KA1-Tr/PETRO-QUANT4<Be>). In case of line overlaps that exceed the limit given in the evaluation model (e.g. by high Pb amounts that cause more than 25 % of the As KA1/Pb LA1 signal), SPECTRA^{plus} automatically activates the alternate line for evaluation (here As KB1/PETRO-QUANT4<Be>). As the sensitivity of a KB1 line is substantially lower than that of a KA1 line, it is advantageous to have this selection done only when needed. In case of L lines, the sensitivity difference is not that significant but the number of lines and thus the number of potential line overlaps are more critical.

Note



As all applications are based on the same measurement method, an erroneous selection of the application can easily be resolved. The sample can be re-evaluated assigning the correct application without running a new measurement. The SPECTRA^{plus} manual will help you to find out how to do this.

7.2 PETRO-QUANT 4<Be> BASIC Analysis of Samples with High Volatility

Samples having a high volatility are analyzed typically by means of the Helium mode of atmospheric pressure ("HeAtm") and Mylar® film.

PETRO-QUANT4<Be>-Fuel defines CH₂ as matrix compound (for samples with considerable oxygen contents please see chapter [PETRO-QUANT 4<Be> BASIC Analysis of Samples with Oxygen \[21\]](#)), a Helium measurement mode of atmospheric pressure and the usage of 3.6 µm Mylar® film.

Of course all automatic line selection features of SPECTRA^{plus} version 2.4.0.4 or higher are integrated in this application as well as in PETRO-QUANT4<Be>-Oil.

7.3 PETRO-QUANT 4<Be> BASIC Analysis of Samples with Oxygen

The application “PETRO-QUANT4<Be> Oil Sulfonatic” additionally takes into account little oxygen contents calculated by stoichiometry from the sulfonatic sulfur content. The Conostan S21 standards with their sulfonatic composition are one example for oil samples that require such an evaluation. Lubricants with high concentrations of esters or used oils with significant amounts of water or alcohols require a compensation of the oxygen's absorption.

The applications “QC<Be>-XRS1 PETRO-QUANT4<Be>” and “QC<Be>-XRS2 PE-PETRO-QUANT4<Be>” consider the oxygen content of two QC samples (for details see table [Table 6.2 \[18j\]](#)).

The application “PETRO-QUANT4<Be> Oil Aut-O-matic” takes into account a significant oxygen content of the analyzed sample. As the application “PETRO-QUANT4<Be>-Oil” sets CH₂ as matrix compound, high oxygen concentrations lead to less accurate results if not considered. As oxygen absorbs the fluorescence radiation more than the matrix compound CH₂, an evaluation ignoring the oxygen content will result in deviations towards smaller concentrations of the other elements. SPECTRA^{plus} allows determining the oxygen content by the Compton coefficient.

The PETRO-QUANT 4<Be> BASIC evaluation model “PETRO-QUANT4<Be> Aut-O-matic” allows having the oxygen content determined by the Compton coefficient. The respective calculations require the consideration of a third preparation parameter. The preferable parameter combination is mass | diameter | density of the specimen. Consequentially, the sample input format “PETRO-QUANT4<Be>.def” allows to input individual values for density and sample mass that differ from the default parameters of the preparation definition stored in the database. The inner sample cup diameter and so the sample diameter is fixed to 3.5 cm of the Bruker AXS liquid cups (order no. 7KP1901-8FA).

Note



As the calculated oxygen content is determined via the Compton coefficient, it is recommended to take care for every parameter affecting it, i.e. sample mass and sample density. The more accurate these parameters are defined, the better is the quality of the analysis results.

When using a modified model (not part of the standard installation) for highly volatile samples (that applies the HeAtm mode and Mylar[®] film), it is recommended to keep the sample mass at 7.0 g.

8 PETRO-QUANT 4 Drift Correction

The PETRO-QUANT 4 drift correction strategy is the following:

The PETRO-QUANT 4<Be> BASIC methods as well as the norm compliant applications that do not use an internal standard, are drift corrected on base of oil samples included in delivery (see chapter [Drift Correction \(Sensitivity\)](#) [23]).

The applications using an internal standard for calibration (ASTM-D5059C, DIN norms DIN13723, DIN 51363, DIN 51391 Ca & Zn, and DIN 51431 as well as ISO14596 (L & H), ISO14597, and ISO15597 need no additional measure of drift correction. The internal standard procedure compensates for instrument drift effects as well as for matrix effects. Any instrument based intensity loss will reduce equally the analyte line and the internal standard line. Therefore, their ratio and consequently the result show no influence of instrument drift.

The decision when to do a drift correction should be based on the results of the measurements of the respective QC samples. The outstanding stability of the S8 TIGER 2nd Generation spectrometers ensures that the PETRO-QUANT 4 drift corrections should not be necessary for weeks to months.

In general, there are two different tasks the PETRO-QUANT4 drift correction procedure takes care of:

The first one is the compensation of the instrument's sensitivity drift. This is the classical drift correction that compensates the aging effects of the tube and the detectors and that is based on samples with sufficient concentrations of the reference elements. This procedure compensates changes in the sensitivity (slope) of the respective calibrations (see chapter [Drift Correction \(Sensitivity\)](#) [23]). The respective DC samples are the same for all modes. This is intuitive for glass samples. Unexpectedly and counter-intuitively the PETRO-QUANT4<Be>-Fuel drift correction procedure (applying the HeAtm mode and 3.6 µm Mylar® film for analysis samples) requires the PETRO-QUANT4<Be>-Oil preparation procedure (applying the He mode of the calibration, i.e. 7 g and 4 µm Prolene® film, for drift correction). The compensation of the film absorption difference is done by the evaluation program.

The second one is the compensation of scattered radiation of film contaminants and suboptimal background modeling of a given sample type and preparation, i.e. subtraction of a blank sample's intensity. The blank correction uses the blank samples "BLANK@..." and hence compensates changes in the offsets of the respective calibrations (see chapter 8.2). Thus, the blank samples of both He modes have to be prepared according to their individual preparation definition.

In case of solid drift correction samples, SPECTRA^{plus} allows to increase the measurement time for drift correction measurements by a factor defined in the external computer's loader program. In case of the norm methods, it is not necessary to prolong the measurement time as the concentrations are set accordingly. In case of the drift correction of PETRO-QUANT 4<Be> BASIC or DIN51399 an increased measurement time is not applicable due to the limited lifetime of the irradiated film of the liquid cups. Thus, SPECTRA^{plus} version 2.4.0.4 or higher introduces a feature to use the average of multiple drift correction measurements. The number of drift correction duplicates are already set and can be modified separately for blank and drift correction samples in **MeasurementSetup** option "Blank and average samples" to "2" for blank samples and "1" for averaged drift correction samples.

Each blank sample (drift) correction with a reference sample name "BLANK@PreparationName" creates two extra blank correction samples in the loader. The respective sample names have the extensions "!1" and "!2". The first sample without extension creates the line library entry and the second and third sample modify the intensity entry to the average of all measurements.

Each drift correction with a reference sample name starting with "AVG@" creates one extra drift correction sample in the loader. The respective sample name has an extension "!1". The first sample without extension creates the line library entry and the second sample modifies the intensity entry to the average of both measurements.

Note



The measurement order has to be the same than the setting in the loader, i.e. the sample without extension has to be measured first followed by the ones with extension in order of increasing numbers "1", "2" etc..
In case of line library maintenance work that includes the re-creation of drift correction entries by means of the **Launcher** option [Batch Quanti.] this order has to be followed as well.

All in all, the following preparation parameters have to be used for the PETRO-QUANT 4<Be> BASIC DC:

Note



Starting the drift correction of the given application will create DC samples in the loader that have to be prepared as described below:

Application	DC Sample S21+K+Sb -500 ppm (Specimens)	Preparation
PETRO-QUANT4<Be> Oil	AVG@PQ4<Be> S21&K&Sb-500ppm (2)	7 g, 4 µm Prolene®
PETRO-QUANT4<Be> Fuel	AVG@PQ4<Be> S21&K&Sb-500 ppm (2)	7 g, 4 µm Prolene®

Application	BLANK Sample Paraffin (Specimens)	Preparation
PETRO-QUANT4<Be> Oil	BLANK@PETRO-QUANT4<Be>-Oil (3)	7 g, 4 µm Prolene®
PETRO-QUANT4<Be> Fuel	BLANK@PETRO-QUANT4<Be>-Fuel (3)	7 g, 3.6 µm Mylar®

The success of any drift correction should be validated by QC measurements (see chapter [PETRO-QUANT 4 Validation and Quality Check Procedure \[15\]](#)). If the quality checks still fail, please, refer to [Appendix 2: Acceptance or Quality Check Troubleshooting \[29\]](#).

Note



We strongly recommend that you make a complete back up copy after validated drift corrections.

8.1 Drift Correction (Sensitivity)

The PETRO-QUANT 4 drift correction is based on the drift correction reference samples which are included in delivery. Every PETRO-QUANT 4 application can be selected to start reference sample measurements with the option "Drift Correction" of the **Loader** program or the touchscreen. The SPECTRA^{plus} manual or your training documents will guide you how to do this.

Method / Application	Preparation	Drift Correction Sample
PETRO-QUANT4<Be>-Oil	PETRO-QUANT4<Be>-Oil	AVG@PQ4<Be>-S21&K&Sb-500ppm (!1)
PETRO-QUANT4<Be>-Fuel	PETRO-QUANT4<Be>-Oil	AVG@PQ4<Be>-S21&K&Sb-500ppm (!1)
ASTM-D2622L	ASTM-D2622L	DC_ASTM-D2622L
ASTM-D2622L-D	ASTM-D2622L-D	DC_ASTM-D2622L-D
ASTM-D2622L-G	ASTM-D2622L-G	DC_ASTM-D2622L-G
ASTM-D2622M	ASTM-D2622M	DC_ASTM-D2622M
ASTM-D2622H	ASTM-D2622H	DC_ASTM-D2622H

Method / Application	Preparation	Drift Correction Sample
ASTM-D6443	ASTM-D6443	DC_ASTM-D6443
DIN51399	DIN51399	AVG@DC-DIN51399 & AVG@DC2-DIN51399
DIN51790	DIN51790	DC_DIN51790
ISO20884L or ISO20884H *	ISO20884	DC_ISO20884

* As both versions of ISO20884 use the same line, it is sufficient to run the drift correction for one version only even if both versions are installed.

Table 8.1: PETRO-QUANT 4 Drift Correction Samples' Measurement

8.2 Drift Correction (Blank Samples)

All PETRO-QUANT 4 methods which use a blank sample are blank drift corrected based on an oil sample included in the delivery. Respective PETRO-QUANT 4 applications can be selected to start the measurements of reference samples with the option "Drift Correction". The SPECTRA^{plus} manual or your training documents will guide you how to do this.

Method / Application	Preparation	Blank Drift Correction Sample
PETRO-QUANT4<Be>-Oil	PETRO-QUANT4<Be>-Oil	BLANK@PETRO-QUANT4<Be>-Oil (!1, !2)
PETRO-QUANT4<Be>-Fuel	PETRO-QUANT4<Be>-Fuel	BLANK@PETRO-QUANT4<Be>-Fuel (!1, !2)
ASTM-D2622L	ASTM-D2622L	BLANK@ASTM-D2622L (!1, !2)
ASTM-D2622L-D	ASTM-D2622L-D	BLANK@ASTM-D2622L-D (!1, !2)
ASTM-D2622L-G	ASTM-D2622L-G	BLANK@ASTM-D2622L-G (!1, !2)
ASTM-D5059C	ASTM-D5059C	BLANK@ASTM-D5059C (!1, !2)
DIN13723	DIN13723	BLANK@DIN13723 (!1, !2)
DIN51399	DIN51399	BLANK@DIN51399 (!1, !2)
ISO20884L or ISO20884H *	ISO20884	BLANK@ISO20884 * (!1, !2)

* As both versions of ISO20884 use the same line, it is sufficient to run the drift correction for one version even if both versions are installed.

Table 8.2: PETRO-QUANT 4 Blank Drift Correction Samples' Measurement

9 Customizing PETRO-QUANT 4<Be> BASIC

The default PETRO-QUANT 4<Be> BASIC is a perfect universal analysis system for a wide variety of petrochemical matrices and samples. However, for specific analytical tasks there may be requirements for:

- Lower Limits of Detection and Counting Times for specific elements.
- Specific combinations of preparation and measurement mode
- Specific calibrations for elements of the PETRO-QUANT 4<Be> BASIC method.
- Methods to analyze additional elements or concentration ranges.

Note



SPECTRA^{plus} is a powerful tool. Modifications of solutions require a thorough understanding of its functionality and structure. If you feel uncomfortable with the possible risks of such advanced features, Bruker AXS offers so called “Application Support” packages to assist you in the creation of customized solutions.

9.1 Lower Limits of Detection and Counting Times

The counting times used in PETRO-QUANT 4<Be> BASIC have been selected to produce best accuracy for the analysis of 32 elements within less than ~17 minutes total analysis time for the S8 TIGER 4kW/3kW or ~23 minutes for the S8 TIGER 1kW in various petrochemical matrices. For some samples it may be desirable to achieve shorter analysis times or lower detection limits. The latter can be done by increasing the counting times of the respective lines in the PETRO-QUANT 4<Be> BASIC measurement method (thus reducing statistical error and therefore improving LOD). Please refer to the SPECTRA^{plus} user manual to understand how to do this.

NOTICE

Attention

The film materials for preparation of the liquid cups have a limited lifetime when being irradiated by X-rays. Therefore, we recommend not to exceed a total measurement time of 30 minutes.

The requirement to prolong the measurement time for some element lines but not to exceed a 30 minutes measurement time might seem unsolvable at times. In this case the list of elements and lines can be reduced. For example, for specific analytical tasks the concentration of S and Cl is roughly known so that either the trace or the major line can be omitted. Furthermore, there might be some elements in the PETRO-QUANT 4<Be> BASIC method that can be deleted from the measurement method, since their measurement is not needed.

Note



All elements having an impact on matrix correction (i.e. elements with concentrations of more than several hundred µg/g - e.g. Sulfur) **have** to be measured to achieve a good result. In addition, all overlaps based on the line Rh KA1/PETRO-QUANT4<Be>-Comp are compensated by measured intensities. Therefore, **do not** skip the Compton line!
Nevertheless, the elements only used for correction purposes can be analyzed with very short measuring times.

The measurement order of the elements is based on potential decomposition and sedimentation risks. To keep this order it is important to start with PETRO-QUANT4<Be>-Oil.mm or with PETRO-QUANT4<Be>-Fuel.mm as these methods contain all lines and order information.

Note



It is strongly recommended to open the existing measurement method and to immediately create a copy by saving it with a different name. Subsequently, work with the copy only. Nevertheless, if the base evaluation model is damaged, it can be retrieved from your last back up or the PETRO-QUANT 4.X Setup CD.

When a new measurement method is created, a respective application file has to be set up as well to allow routine analysis (see next chapter [Customized PETRO-QUANT 4<Be> BASIC Applications \[► 26\]](#)).

9.2 Customized PETRO-QUANT 4<Be> BASIC Applications

There are a few situations that make it necessary to modify PETRO-QUANT 4<Be> BASIC applications:

In case new measurement methods have been created, the respective applications have to be set up for routine analysis functionality.

For some real samples the default combination of low volatile samples and Prolene® film preparation may turn out to be insufficient. In case of unknown samples it is worth testing the resistance of the film material before the analysis (see note at the end of chapter [Sample Preparation in Liquid Cups \[► 6\]](#)).

The best way to create a new evaluation model is to modify the one that is most similar to one that already exists by means of **ApplicationSetup**.

Note



It is strongly recommended to open the existing application and to immediately create a copy by saving it with a different name. Subsequently, work with the copy only. Nevertheless, if the base evaluation model is damaged, it can be retrieved from your last back up or the PETRO-QUANT 4.X Setup CD.

The new evaluation model now can be used to define the appropriate combination of preparation, measurement method (and thus measurement mode), matrix compound, and selection of lines used for automatic evaluation. The SPECTRA^{plus} manual describes how to do this.

Once the application is created, it can be used for drift correction - initially, if new mode-blank combinations have been set, and later for routine analysis.

9.3 Customized Calibration Methods

The line library delivered with your S8 TIGER system contains analytical lines specifically optimized for trace element analysis of petrochemical materials. These lines are automatically used in the two default applications – PETRO-QUANT4<Be>-Oil and PETRO-QUANT4<Be>-Fuel – but they can also be used in the creation of your own calibration methods.

For some specialized analytical tasks, the direct application of PETRO-QUANT 4<Be> BASIC may lack sufficient accuracy or precision (see Sulfur example below). When creating a new calibration with your own standard samples, it is possible to select the PETRO-QUANT4<Be> lines for your measurement method. This saves you the task of creating your own lines and spending time working out instrument conditions, peak/background positions, pulse height discrimination and drift correction settings. To learn more about creating new calibrations, please refer to the SPECTRA^{plus} user manual.

The analysis of lowest µg/g ranges of S shows a LoD of about 1-2 mg/kg applying PETRO-QUANT4-Oil directly. Setting up a one-element calibration based on a few standards from 0 to 20 mg/kg reduces the LoD to 0.3 mg/kg applying a measurement time of 100 s for peak and background positions. This can be done in less than one hour, preparation of standard samples excluded.

Note



The PETRO-QUANT 4<Be> BASIC measurement parameters are optimized for the multi-elemental analysis in different matrices. Therefore, the lines have up to four background positions which, to some extent, are shared amongst several elements. Customer specific application with a limited variety of matrices and number of elements may allow creating a copy of the respective line of PETRO-QUANT 4<Be> BASIC to delete some of the background positions and so to save measurement time

9.4 Additional Elements or Expanded Concentration Ranges

PETRO-QUANT 4<Be> BASIC will handle all samples that fulfill the requirements presented in the second paragraph of chapter [Introduction \[5 \]](#). In case of entirely unknown samples, this match can be checked easily by the Bruker AXS QUANT-EXPRESS system (optionally available with the S8 TIGER 2nd Generation). After checking the appropriate preparation technique, the next step in the investigation of an unknown sample should be the standardless analysis of the sample. As this step is meant to be an overview analysis, a fast measurement method can be applied. This is sufficient to find out:

- Is the matrix description “CH₂” appropriate? An insufficient matrix description is indicated in the interactive evaluation by the Compton coefficient which in e.g. PETRO-QUANT4<Be>-Oil should be very close (± 0.05) to 1.00. A value higher than 1.05 indicates that the matrix contains heavier elements like oxygen (or additional elements not included in the PETRO-QUANT 4<Be> BASIC method). The application “PETRO-QUANT4<Be> Oil Aut-O-matic” is set up to determine this oxygen content from the Compton coefficient deviation.

In this first case, the matrix is automatically optimized. If the respective sample type is expected to be analyzed frequently, it is recommended to create a new application with fixed oxygen content (see chapter [Customized PETRO-QUANT 4<Be> BASIC Applications \[26 \]](#)).

- Is any element that is not covered by PETRO-QUANT 4<Be> BASIC present in a concentration range that causes matrix effects or line overlay interferences?
- Is any element included in PETRO-QUANT 4<Be> BASIC present in a concentration much higher than described in chapter [Introduction \[5 \]](#) (so that the measurement parameters created for a trace (all) or minor (S and Cl) analysis are not applicable)?

In these latter two cases, the very powerful tool SPECTRA^{plus} offers a simple way to create expanded analytical methods by mixing PETRO-QUANT 4<Be> BASIC with the optional QUANT-EXPRESS system.

Standardless analysis as well as PETRO-QUANT 4<Be> BASIC analysis is based on calibrations stored in the line library. So, including lines of the standardless system into a renamed copy of the PETRO-QUANT 4 BASIC measurement method (see chapter [Lower Limits of Detection and Counting Times \[25 \]](#)) and the creation of a respective application (see chapter [Customized PETRO-QUANT 4<Be> BASIC Applications \[26 \]](#)) will allow an expanded analysis with interactive evaluation. These new lines have to be calibrated and drift corrected for the mode that will be used. This can be checked via **LibManager**.

If the new line has to be chosen due to concentration mismatches, i.e. if the element list is not changed, the work is done now.

If the new application contains additional elements and is meant to be used as a routine analysis method, the last step to do is to expand the results window format via **Results Manager** and to address this in the new application method (see chapter [Customized PETRO-QUANT 4<Be> BASIC Applications](#) [▶ 26]). Please refer to the SPECTRA^{plus} manual to remember how to do these additional steps.

Note

The procedure described above will help to improve matrix correction by taking new elements into account or by considering more accurate amounts of elements present in a concentration much higher than given in chapter [Introduction](#) [▶ 5].

It remains the responsibility of the analyst to check for possible interferences of these compounds not foreseen in the design of PETRO-QUANT 4<Be> BASIC. Any new element may not only affect the matrix correction but also lead to line overlay and so bias the result of another (mostly adjacent) element. In this case even replace the disturbed PETRO-QUANT 4<Be> BASIC line by one of the standardless system as this one will consider the respective line overlays

It is important to know that the measurement mode itself (scan or peak/background) has no influence on the evaluation. Any PETRO-QUANT 4<Be> BASIC line set to scan mode will work as well as any calibrated copy of a QUANT-EXPRESS line set to peak/background mode. If the new line belongs to the standardless system, the counting statistical deviation can be improved by creating a calibrated copy to define peak and background positions. After modification it can be measured in peak/background mode as well as the PETRO-QUANT 4<Be> BASIC lines.

This SPECTRA^{plus} way of evaluations is based on measurement methods that are made up from calibrated lines only. Any user specific mixture of PETRO-QUANT 4<Be> BASIC and the standardless package QUANT-EXPRESS can be compiled easily - an unrivalled feature of the Bruker AXS software.

9.5 Partial Calibrations for Additional Elements

If the accuracy and/or precision of the line imported from the standardless system QUANT-EXPRESS should be improved even further, SPECTRA^{plus} allows you to use partial calibrations. This feature is activated by following just a few simple steps. You can create special calibrations with standard samples for the additional elements only. To do so, a SPECTRA^{plus} material has to be set up that contains all required (by customer and matrix correction requests) PETRO-QUANT 4<Be> BASIC elements and the new elements. It is not necessary that the standard samples contain the PETRO-QUANT 4<Be> BASIC elements. So, it would be sufficient to have a set of 5 to 8 standard samples of known composition for any additional element. The calibration software of SPECTRA^{plus} allows suppressing the automatic calculation of new calibration parameters. If you are in doubt how to do so, please refer to the SPECTRA^{plus} manual. When creating the calibration, the PETRO-QUANT 4<Be> BASIC element lines' calibration parameters will be retrieved from the line library. They are kept whereas the calibration for the new elements will be based on the standard samples' measurements.

Thus, depending upon the number of elements kept from PETRO-QUANT 4<Be> BASIC and hence on the measurement time for the new standard samples it just takes 1-2 hours per element to compile specific elements' calibrations with the PETRO-QUANT 4<Be> BASIC system in the best quality of specialized XRF methods.

10 Appendix

10.1 Appendix 1: Back-Up and Re-Installation of the PETRO-QUANT 4 Method

Your S8 TIGER 2nd Generation is delivered with PETRO-QUANT 4 factory installed and all of the software files are optimized and located in the correct directories.



Note

We strongly recommend that you make regular back-up copies of all key libraries, databases and files that you use routinely with your S8 system.

Your SPECTRA^{plus} manual contains information regarding data back-up. With regular back-ups it will be straightforward to retrieve analytical data and methods from PETRO-QUANT 4.

However, if there is a problem with your spectrometer PC that - due to lack of a back-up copy - requires a brand new installation of SPECTRA^{plus}, it may be necessary that the PETRO-QUANT 4 package also needs to be re-installed.

Please contact your local Bruker AXS service team for support if you suspect that you need a new installation of PETRO-QUANT 4.

10.2 Appendix 2: Acceptance or Quality Check Troubleshooting

The PETRO-QUANT 4 package was fully tested and checked before it left the factory. If there are problems with the validation procedure, an error must have occurred either during transportation, during the installation or with the condition of the check samples.

Listed below are some suggestions for troubleshooting.

Acceptance Samples Mix-Up

The acceptance samples are all delivered in the same kind of HDPE or glass bottles. If you find a discrepancy running your quality check (QC) samples, please, check that QC sample and selected application match.



Note

Do never pour back samples to the original bottles!
Some of the validation samples can be measured for setup purposes again, but we strongly recommend storing used material separately from the original samples.

Condition of the Acceptance Samples

Liquid samples have an expiration date. Typical values of stabilized commercial standard samples are 1-3 years unopened and 1-2 years opened. The lifetime of non-stabilized samples may be reduced to weeks and for very low concentrations these samples have to be used on the day of preparation.

If deviations occur running the QC samples, please, check the age of the samples. Furthermore, it is worth to perform a visual check for precipitation or the formation of layers. If it is to be assumed that a QC sample has been intermixed, it can be checked by a full analysis with PETRO-QUANT 4<Be> BASIC.

Drift Correction (DC)

The intensities collected from the DC samples are used to optimize your PETRO-QUANT 4 results.

During the installation of PETRO-QUANT 4 a complete set of drift correction measurements is run. If there was a problem with the analysis of these DC samples, this may cause errors with your data. If in the course of time the QC samples run out of specification limits and the samples themselves are visually OK and not intermixed (see last two sections), a drift correction is the first measure of correction.

If only the blank sample QC measurement fails, it is sufficient to run the blank drift correction. If the blank sample is OK but at least one of the other QC samples fails, it may be sufficient to run the AVG@sample only. However, we recommend running a complete drift correction set each time.

After finishing the above steps, you can re-evaluate and re-run your QC samples and compare the match of the values with the specifications indicated by the color markers of the results display.

Condition of the Spectrometer

If the QC measurements still show mismatches with the tolerances of the specifications after drift correction, the condition of the spectrometer should be checked.

During the spectrometer installation, a full system alignment has been run. The alignment checks for accuracy of the goniometer, the optimum detector voltages, and the collimator alignment. If any of these parameters are not set correctly, this may produce errors of your PETRO-QUANT 4 applications.

If necessary the alignment procedure can be carried out again using **SpectrometerAlignment**. The SPECTRA^{plus} manual will tell you how to use this program.

Once the alignment has been finished, the PETRO-QUANT 4 drift correction procedure should be carried out again (refer to chapter [PETRO-QUANT 4 Drift Correction \[22\]](#)) and the QC samples should be re-run.

Peak Position Check

If the QC samples' mismatch remains even after spectrometer alignment and drift correction, it is worth to check peak position and background positions. The peak and background positions for the PETRO-QUANT 4 lines should be automatically correct once the **SpectrometerAlignment** is finished successfully. However, if accuracy problems persist, it is possible to re-align the positions. This requires a new drift correction followed by QC measurements.

10.3 Appendix 3: PETRO-QUANT 4<Be> BASIC Reference Samples

	Na [mg/kg]	Mg [mg/kg]	Al [mg/kg]	Si [mg/kg]	P [mg/kg]	S [%]	Cl [mg/kg]
QC<Be>-Paraffin PETRO-QUANT4<Be>	0	0	0	0	0	0.0000	0
Alignment<-Be>-CON S21&K&Sb 500 ppm	500	500	500	500	500	[1.28]	
Alignment<-Be>-XRS1 PETRO-QUANT4<Be>	900					0.0500	10000
Alignment<-Be>-XRS2 PETRO-QUANT4<Be>						[1.24]	
QC<Be>-CON S21&K&Sb 100 ppm	100	100	100	100	100	[0.255]	
QC<Be>-XRS1 PETRO-QUANT4<Be>						0.0100	100
QC<Be>-XRS2 PETRO-QUANT4<Be>						[0.058]	
[] informational values							

	K [mg/kg]	Ca [mg/kg]	Ti [mg/kg]	V [mg/kg]	Cr [mg/kg]	Mn [mg/kg]	Fe [mg/kg]
QC<Be>-Paraffin PETRO-QUANT4<Be>	0	0	0.0	0.0	0.0	0.0	0.0
Alignment<-Be>-CON S21&K&Sb 500 ppm	500	500	500.0	500.0	500.0	500.0	500.0
Alignment<-Be>-XRS1 PETRO-QUANT4<Be>							
Alignment<-Be>-XRS2 PETRO-QUANT4<Be>							
QC<Be>-CON S21&K&Sb 100 ppm	100	100	100.0	100.0	100.0	100.0	100.0
QC<Be>-XRS1 PETRO-QUANT4<Be>							
QC<Be>-XRS2 PETRO-QUANT4<Be>							

PETRO-QUANT
For S8 TIGER 2nd Generation
User Manual



	Co [mg/kg]	Ni [mg/kg]	Cu [mg/kg]	Zn [mg/kg]	As [mg/kg]	Br [mg/kg]	Zr [mg/kg]
QC<Be>-Paraffin PETRO-QUANT4<Be>	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Alignment<-Be>-CON S21&K&Sb 500 ppm		500.0	500.0	500.0			
Alignment<-Be>-XRS1 PETRO-QUANT4<Be>						350.0	350.0
Alignment<-Be>-XRS2 PETRO-QUANT4<Be>	250.0				350.0		
QC<Be>-CON S21&K&Sb 100 ppm		100.0	100.0	100.0			
QC<Be>-XRS1 PETRO-QUANT4<Be>						100.0	100.0
QC<Be>-XRS2 PETRO-QUANT4<Be>	100.0				100.0		

	Mo [mg/kg]	Ag [mg/kg]	Cd [mg/kg]	In [mg/kg]	Sn [mg/kg]	Sb [mg/kg]	Ba [mg/kg]
QC<Be>-Paraffin PETRO-QUANT4<Be>	0.0	0	0	0	0	0	0
Alignment<-Be>-CON S21&K&Sb 500 ppm	500.0	500	500		500	500	500
Alignment<-Be>-XRS1 PETRO-QUANT4<Be>							
Alignment<-Be>-XRS2 PETRO-QUANT4<Be>				500			
QC<Be>-CON S21&K&Sb 100 ppm	100.0	100	100		100	100	100
QC<Be>-XRS1 PETRO-QUANT4<Be>							
QC<Be>-XRS2 PETRO-QUANT4<Be>				100			

PETRO-QUANT
For S8 TIGER 2nd Generation
User Manual



	W [mg/kg]	TI [mg/kg]	Pb [mg/kg]	Bi [mg/kg]	CH ₂ [%]	O [%]	ρ [g/cm ³]
QC<Be>-Paraffin PETRO-QUANT4<Be>	0.0	0.0	0.0	0.0	100.0000	0.0	0.86
Alignment<-Be>-CON S21&K&Sb 500 ppm			500.0			[1.91]	0.90
Alignment<-Be>-XRS1 PETRO-QUANT4<Be>						1.03	0.87
Alignment<-Be>-XRS2 PETRO-QUANT4<Be>	500.0	250.0		250.0		0.334	0.87
QC<Be>-CON S21&K&Sb 100 ppm			100.0			[0.382]	0.88
QC<Be>-XRS1 PETRO-QUANT4<Be>						1.00	0.87
QC<Be>-XRS2 PETRO-QUANT4<Be>	100.0	100.0		100.0		0.088	0.87

10.4 Appendix 4: Oil Density Determination

Sample Mass* [g]	Density [g/cm ³]	Sample Mass* [g]	Density [g/cm ³]	Sample Mass* [g]	Density [g/cm ³]
2.46 2.47 2.48 2.49	0.65	2.50 2.51 2.52 2.53	0.66	2.54 2.55 2.56 2.57	0.67
2.58 2.59 2.60	0.68	2.61 2.62 2.63 2.64	0.69	2.65 2.66 2.67 2.68	0.70
2.69 2.70 2.71 2.72	0.71	2.73 2.74 2.75 2.76	0.72	2.77 2.78 2.79 2.80	0.73
2.81 2.82 2.83	0.74	2.84 2.85 2.86 2.87	0.75	2.88 2.89 2.90 2.91	0.76
2.92 2.93 2.94 2.95	0.77	2.96 2.97 2.98 2.99	0.78	3.00 3.01 3.02	0.79
3.03 3.04 3.05 3.06	0.80	3.07 3.08 3.09 3.10	0.81	3.11 3.12 3.13 3.14	0.82
3.15 3.16 3.17 3.18	0.83	3.19 3.20 3.21	0.84	3.22 3.23 3.24 3.25	0.85
3.26 3.27 3.28 3.29	0.86	3.30 3.31 3.32 3.33	0.87	3.34 3.35 3.36 3.37	0.88
3.38 3.39 3.40	0.89	3.41 3.42 3.43 3.44	0.90	3.45 3.46 3.47 3.48	0.91
3.49 3.50 3.51 3.52	0.92	3.53 3.54 3.55 3.56	0.93	3.57 3.58 3.59 3.60	0.94
3.61 3.62 3.63	0.95	3.64 3.65 3.66 3.67	0.96	3.68 3.69 3.70 3.71	0.97
3.72 3.73 3.74 3.75	0.98	3.76 3.77 3.78 3.79	0.99	3.80 3.81 3.82	1.00
3.83 3.84 3.85 3.86	1.01	3.87 3.88 3.89 3.90	1.02	3.91 3.92 3.93 3.94	1.03
3.95 3.96 3.97 3.98	1.04	3.99 4.00 4.01	1.05	4.02 4.03 4.04 4.05	1.06
4.06 4.07 4.08 4.09	1.07	4.10 4.11 4.12 4.13	1.08	4.14 4.15 4.16 4.17	1.09
4.18 4.19 4.20 4.21	1.10	4.22 4.23 4.24	1.11	4.25 4.26 4.27 4.28	1.12
4.29 4.30 4.31 4.32	1.13	4.33 4.34 4.35 4.36	1.14	4.37 4.38 4.39 4.40	1.15
4.41 4.42 4.43	1.16	4.44 4.45 4.464 4.47	1.17	4.48 4.49 4.50 4.51	1.18
4.52 4.53 4.54 4.55	1.19	4.56 4.57 4.58 4.59	1.20	4.60 4.61 4.62	1.21
4.63 4.64 4.65 4.66	1.22	4.67 4.68 4.69 4.70	1.23	4.71 4.72 4.73 4.74	1.24
4.75 4.76 4.77 4.78	1.25	4.79 4.80 4.81	1.26	4.82 4.83 4.84 4.85	1.27
4.86 4.87 4.88 4.89	1.28	4.90 4.91 4.92 4.93	1.29	4.94 4.95 4.96 4.97	1.30
4.98 4.99 5.00 5.01	1.31	5.02 5.03 5.04	1.31	5.05 5.06 5.07 5.08	1.32
5.09 5.10 5.11 5.12	1.34	5.13 5.14 5.15 5.16	1.35	5.17 5.18 5.19 5.20	1.36
5.21 5.22 5.23	1.37	5.24 5.25 5.26 5.27	1.38	5.28 5.29 5.30 5.31	1.39
5.32 5.33 5.34 5.35	1.40	5.36 5.37 5.38 5.39	1.41	5.40 5.41 5.42	1.42
5.43 5.44 5.45 5.46	1.43	5.47 5.48 5.49 5.50	1.44	5.51 5.52 5.53 5.54	1.45
5.55 5.56 5.57 5.58	1.46	5.59 5.60 5.61	1.47	5.62 5.63 5.64 5.65	1.48
5.66 5.67 5.68 5.69	1.49	5.70 5.71 5.72 5.73	1.50	5.74 5.75 5.76 5.77	1.51
5.78 5.79 5.80 5.81	1.52	5.82 5.83 5.84	1.53	5.85 5.86 5.87 5.88	1.54
5.89 5.90 5.91 5.92	1.55	5.93 5.94 5.95 5.96	1.56	5.97 5.98 5.99 6.00	1.57
6.01 6.02 6.03	1.58	6.04 6.05	1.59		

* Sample mass without cup; cup filled to a flat surface at the top, i.e. no meniscus.

11 Applications and Drift Correction

Application (*.evm)	Mode	Preparation	Evaluation	Purpose
PETRO-QUANT4<Be> Oil	Helium reduced	7.0 g 4 µm Prolene®	Matrix CH ₂	Oil Samples without significant amounts of O (< 0.5 %)
PETRO-QUANT4<Be> Oil Aut-O-matic	Helium reduced	7.0 g 4 µm Prolene®	Matrix CH ₂ O by Compton	Oil Samples with significant amounts of O (≥ 0.5 %)
PETRO-QUANT4<Be> Oil Sulfonatic	Helium reduced	7.0 g 4 µm Prolene®	Matrix CH ₂ S as SO ₃	Sulfonatic Oil Samples like CONOSTAN
QC<Be>-CON S21&K&Sb 100 ppm	Helium reduced	7.0 g 4 µm Prolene®	Matrix CH ₂ S as SO ₃	QC Sample CONOSTAN S21+K+Sb 100 ppm
QC<Be>-Paraffin PETRO-QUANT4<Be>	Helium reduced	7.0 g 4 µm Prolene®	Matrix CH ₂	QC Sample Paraffin
QC<Be>-XRS1 PETRO-QUANT4<Be>	Helium reduced	7.0 g 4 µm Prolene®	Matrix CH ₂ 1.00 % O	QC Sample XRS1
QC<Be>-XRS2 PETRO-QUANT4<Be>	Helium reduced	7.0 g 4 µm Prolene®	Matrix CH ₂ 0.088 % O	QC Sample XRS2
PETRO-QUANT4<Be> Fuel	Helium atmospheric	7.0 g 3:6 µm Mylar®	Matrix CH ₂	Highly volatile samples like Gasoline or samples Prolene® is not chemically resistant to

Table 11.1: PETRO-QUANT4<Be> Applications

Application (*.evm)	Sample Name	Mode	Preparation	Comment
PETRO-QUANT4<Be> Oil	AVG@PQ4<Be>S21&K&Sb-500ppm	Helium reduced	7.0 g 4 µm Prolene®	
	BLANK@PETRO-QUANT4<Be>-Oil	Helium reduced	7.0 g 4 µm Prolene®	
PETRO-QUANT4<Be> Fuel	AVG@PQ4<Be>S21&K&Sb-500ppm	Helium reduced	7.0 g 4 µm Prolene®	
	BLANK@PETRO-QUANT4<Be>-Fuel	Helium atmospheric	7.0 g 3:6 µm Mylar®	Only sample for drift correction prepared with Mylar film

Table 11.2: PETRO-QUANT4<Be> Drift Correction



Bruker Corporation

info.baxs@bruker.com
www.bruker.com